UNIVERSITY OF SCIENCE, VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
FACULTY OF INFORMATION TECHNOLOGY
COURSE: INTRODUCTION TO INFORMATION TECHNOLOGY (CSC00004)

PROJECT 2: DEVELOPING AN AI APPLICATION
FIRST SUBMISSION PROGRESS REPORT

Project Topic: 3D Object Reconstruction from 2D Images using TRELLIS.2 (Microsoft)

Team Name:	**ae cây khế**
Class ID / Section:	26C05
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1. Project Overview

1.1. Problem Statement

In digital content creation, game development, e-commerce, and virtual reality, generating detailed 3D models from 2D images is an essential but challenging task. Manual 3D modeling requires substantial time, artistic skill, and specialized software. Automating 3D asset generation directly from single 2D images streamlines digital production pipelines and enables users without 3D design expertise to create interactive spatial assets.

1.2. Proposed AI Solution

The team selected the problem of 3D object reconstruction from 2D images. The project utilizes TRELLIS.2, an advanced 3D generation model developed by Microsoft and hosted on Hugging Face Spaces. The planned application will allow users to upload a 2D image, process it through the AI model, and obtain a standard 3D model file (such as .glb) suitable for interactive web rendering and downloading.

2. Current Progress and Task Distribution

During the initial phase of the project, the team focused on establishing the project topic, defining the core application scope, and distributing major roles among five group members. Technical implementation and code setup have not commenced yet and are scheduled for the next development sprint.

Member	Assigned Role	Planned Responsibilities	Current Status
Member 1: Lê Bá Thi	AI Model & Inference	1. Study TRELLIS.2 repository and setup model inference execution. 2. Test 3D generation from sample 2D images and verify .glb output. 3. Build Python module to process image inputs and return 3D files.	Topic approved, role assigned, technical setup pending.
Member 2: Nguyễn Hồng Bảo Trân	Frontend & UI/UX	1. Build web application interface. 2. Implement image upload panel and processing status indicator. 3. Integrate 3D model viewer and file download interface.	Topic approved, role assigned, UI development pending.
Member 3: Trịnh Duy Khoa	Backend & Integration	1. Design image upload validation and backend routing logic. 2. Build intermediate API connecting frontend with AI inference module. 3. Manage temporary file storage and process handling.	Topic approved, role assigned, backend setup pending.
Member 4: Phạm Tấn Thành	Testing, Evaluation & Demo	1. Prepare diverse test image dataset for 3D reconstruction. 2. Evaluate model output quality and measure processing latency. 3. Document generation artifacts and compile demonstration materials.	Topic approved, role assigned, testing dataset pending.
Member 5: Trần Minh Khoa	Project Coordinator & Documentation	1. Manage project schedule, GitHub repository, and team coordination. 2. Prepare First Submission progress report. 3. Coordinate final report, slide deck, and submission packaging.	Topic approved, progress report completed, repo setup pending.

3. Current Challenges

3.1. External Execution Environment for TRELLIS.2

The main challenge currently facing the team is identifying a suitable external hosting and execution solution for the TRELLIS.2 model. Due to high VRAM and GPU hardware requirements, running inference locally on consumer-grade hardware is inefficient. The team is evaluating external cloud execution options, including Hugging Face Spaces API integration and cloud GPU environments.

3.2. Framework and Technology Stack Selection

The specific technology stack for frontend user interface and backend communication has not been locked down yet. The team needs to compare frameworks like Gradio and Streamlit to determine which tool provides the most seamless integration with 3D model visualization components.

4. Next Steps and Planned Timeline

The team will finalize technology decisions and detailed task breakdowns over the next 2 to 3 days, followed by system implementation in the upcoming weeks.

Timeline	Planned Objectives	Responsible Members
Next 2 to 3 Days	1. Hold team sync to reach consensus on technology stack selection. 2. Finalize detailed task assignments and API specifications for each member. 3. Decide on external cloud execution approach for TRELLIS.2 model.	All 5 Team Members
Week 2 (Implementation)	1. Set up TRELLIS.2 inference workflow on selected cloud platform. 2. Build basic UI prototype with 3D model viewer. 3. Connect frontend UI with model inference backend. 4. Begin assembling test image dataset.	Member 1, Member 2, Member 3, Member 4
Week 3 (Finalization)	1. Perform system testing and quality evaluation on generated 3D assets. 2. Write Final Report document (Final_Report.pdf). 3. Design presentation slide deck (Slide.pdf) and prepare oral presentation. 4. Verify directory structure and package submission ZIP file.	Member 4, Member 5, and All Members